



Solving Systems of Linear Equations by Substitution

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Learning Objectives

- Solve a linear system by isolating one variable and substituting into the other equation
- Identify the solution as an ordered pair representing the point of intersection
- Recognize systems with no solution or infinitely many solutions

Solve each system of linear equations using the substitution method and write your answer as an ordered pair.

1. Solve the system using substitution.

$$\begin{cases} y = 2x + 1 \\ 3x + y = 11 \end{cases}$$

Answer: _____

2. Solve the system using substitution.

$$\begin{cases} x = y + 3 \\ 2x + y = 12 \end{cases}$$

Answer: _____

3. Solve the system using substitution.

$$\begin{cases} y = 3x - 4 \\ 2x + y = 6 \end{cases}$$

Answer: _____

4. Solve the system using substitution.

$$\begin{cases} x + y = 7 \\ 2x - y = 5 \end{cases}$$

Answer: _____

5. Solve the system using substitution.

$$\begin{cases} x = 2y \\ 2x + 3y = 14 \end{cases}$$

Answer: _____

6. Solve the system using substitution.

$$\begin{cases} y = -x + 5 \\ 3x - 2y = 5 \end{cases}$$

Answer: _____



7. Solve the system using substitution.

$$\begin{cases} 4x - y = 1 \\ y = 2x + 3 \end{cases}$$

Answer: _____

8. Solve the system using substitution.

$$\begin{cases} x - 2y = -1 \\ 3x + y = 11 \end{cases}$$

Answer: _____

9. Solve the system using substitution.

$$\begin{cases} y = 4x - 9 \\ y = x + 3 \end{cases}$$

Answer: _____

10. Solve the system using substitution.

$$\begin{cases} 2x + y = 8 \\ x - y = 1 \end{cases}$$

Answer: _____





Encourage students to first isolate the variable with a coefficient of 1 to avoid fractions when substituting.

Solutions

1. Solve the system using substitution.

$$\begin{cases} y = 2x + 1 \\ 3x + y = 11 \end{cases}$$

- Apply the substitution method to the system $[y = 2x + 1]$ and $[3x + y = 11]$.
- Reduce to one variable and solve to get $x = 2$.
- Substitute back to get $y = 5$.
- The solution is $(2, 5)$; it checks in both original equations.

Answer: (2, 5)

2. Solve the system using substitution.

$$\begin{cases} x = y + 3 \\ 2x + y = 12 \end{cases}$$

- Apply the substitution method to the system $[x = y + 3]$ and $[2x + y = 12]$.
- Reduce to one variable and solve to get $x = 5$.
- Substitute back to get $y = 2$.
- The solution is $(5, 2)$; it checks in both original equations.

Answer: (5, 2)

3. Solve the system using substitution.

$$\begin{cases} y = 3x - 4 \\ 2x + y = 6 \end{cases}$$

- Apply the substitution method to the system $[y = 3x - 4]$ and $[2x + y = 6]$.
- Reduce to one variable and solve to get $x = 2$.
- Substitute back to get $y = 2$.
- The solution is $(2, 2)$; it checks in both original equations.

Answer: (2, 2)

4. Solve the system using substitution.

$$\begin{cases} x + y = 7 \\ 2x - y = 5 \end{cases}$$

- Apply the substitution method to the system $[x + y = 7]$ and $[2x - y = 5]$.
- Reduce to one variable and solve to get $x = 4$.
- Substitute back to get $y = 3$.
- The solution is $(4, 3)$; it checks in both original equations.

Answer: (4, 3)



5. Solve the system using substitution.

$$\begin{cases} x = 2y \\ 2x + 3y = 14 \end{cases}$$

→ Apply the substitution method to the system $[x = 2y]$ and $[2x + 3y = 14]$.

→ Reduce to one variable and solve to get $x = 4$.

→ Substitute back to get $y = 2$.

→ The solution is $(4, 2)$; it checks in both original equations.

Answer: (4, 2)

6. Solve the system using substitution.

$$\begin{cases} y = -x + 5 \\ 3x - 2y = 5 \end{cases}$$

→ Apply the substitution method to the system $[y = -x + 5]$ and $[3x - 2y = 5]$.

→ Reduce to one variable and solve to get $x = 3$.

→ Substitute back to get $y = 2$.

→ The solution is $(3, 2)$; it checks in both original equations.

Answer: (3, 2)

7. Solve the system using substitution.

$$\begin{cases} 4x - y = 1 \\ y = 2x + 3 \end{cases}$$

→ Apply the substitution method to the system $[4x - y = 1]$ and $[y = 2x + 3]$.

→ Reduce to one variable and solve to get $x = 2$.

→ Substitute back to get $y = 7$.

→ The solution is $(2, 7)$; it checks in both original equations.

Answer: (2, 7)

8. Solve the system using substitution.

$$\begin{cases} x - 2y = -1 \\ 3x + y = 11 \end{cases}$$

→ Apply the substitution method to the system $[x - 2y = -1]$ and $[3x + y = 11]$.

→ Reduce to one variable and solve to get $x = 3$.

→ Substitute back to get $y = 2$.

→ The solution is $(3, 2)$; it checks in both original equations.

Answer: (3, 2)

9. Solve the system using substitution.

$$\begin{cases} y = 4x - 9 \\ y = x + 3 \end{cases}$$

→ Apply the substitution method to the system $[y = 4x - 9]$ and $[y = x + 3]$.

→ Reduce to one variable and solve to get $x = 4$.

→ Substitute back to get $y = 7$.

→ The solution is $(4, 7)$; it checks in both original equations.

Answer: (4, 7)



10. Solve the system using substitution.

$$\begin{cases} 2x + y = 8 \\ x - y = 1 \end{cases}$$

→ Apply the substitution method to the system $[2x + y = 8]$ and $[x - y = 1]$.

→ Reduce to one variable and solve to get $x = 3$.

→ Substitute back to get $y = 2$.

→ The solution is $(3, 2)$; it checks in both original equations.

Answer: $(3, 2)$

